

Elouan Salaun

Data Scientist | Machine Learning Engineer

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PROFILE

Trained as a generalist engineer and data scientist, I build decision-support systems end to end: data collection and NLP extraction, predictive modelling, deployment on Azure, and adoption by business users. My last work lifted the simulated win-rate of a bid-pricing process from roughly 15% to 40% and projects a 62% increase in expected gains. Targeting a new opportunity to apply the mathematical and scientific rigor developed through my engineering education and research background.

EXPERIENCE

Data Science Research Intern | Commissionnaires du Québec

Montréal, sept.2024 – sept.2026

Security-services organization bidding on public tenders across Québec. Sole owner of the data & AI workstream, from scoping to deployment.

- Designed and shipped a complete AI-driven tool covering the full RFP response process, now in sustained daily use by bid managers; presented multiple times to the executive committee and to all staff.
- Built a pricing model (XGBoost / Learning-to-Rank) combining market dynamics with contract-specific technical features: simulated win rate of 40% vs. 15% for the pricing rule in use, +62% projected gains and +22% against historical results on retrospective backtests.
- Consolidated the data foundation from scratch (internal spreadsheets and PDFs plus external APIs and scraping) into a single contract database; extraction from contractual documents via NLP and hierarchical RAG.
- Developed a similar-contract retrieval model (neural network with human feedback) to surface comparable past bids for any new tender.
- Delivered the decision-support application (Streamlit) and deployed the stack on Azure: ADLS Gen2 data lake, Functions, AppService, Containers.
- Built automated Excel spreadsheets with macros (VBA) for reporting outside the tool.

PUBLICATION

Leveraging artificial intelligence and strategic data for tender response processes

CENTERIS 2026

E. Salaun, M. Belmouadden, R. Pellerin, A. Neumann – conference paper, Procedia Computer Science.

SELECTED PROJECTS

SEAO Public Procurement MCP Connector

2026

- Local MCP server that lets an LLM interrogate Québec's SEO open procurement data in plain language: ingestion pipeline from raw publications, cleaning and normalization into SQLite, hybrid query core, concurrent analysis inside the model's reasoning loop.

Open source: github.com/elouansalaun/SEO_MCP

Investment portfolio optimization

2025

- Liability-driven portfolio construction: Markowitz frontier and GMV under estimation-error sensitivity, Gaussian and Cornish-Fisher VaR/CVaR, CPPI with explicit drawdown constraint, geometric Brownian motion Monte Carlo, CIR rates and bond pricing.

Behavioral Analysis of a Superalloy | Safran Aircraft Engines – semester project

2024

- Instrumented a forging process, then identified the constitutive law from the collected data using machine learning; validated against numerical simulations.

Rocket Nose Cone Manufacturing by Filament Winding | CNES (Perseus)– semester project

2024

- Determined filament lay-down trajectories as geodesics, programmed the 4-axis winding machine (G-code), and built an interface to compile, verify and visualize the trajectories.

Connected Multi-Point Cash Register System | ENSAM Angers Grand Gala

2023

- Designed and programmed an ESP32 real-time sales-recording device for a 3,000-person event: MQTT transmission, centralized inventory and demand forecasting with restocking anticipation across points of sale.

EDUCATION

Research Master's (MSc.A), Industrial Engineering | Polytechnique Montréal

Montréal, 2024-2026

Thesis: *Tender response process: integrating artificial intelligence methods that leverage contractual and competitive data.*

Supervised by Robert Pellerin and Anas Neumann

Member of: Jarislowsky/AtkinsRéalis research chair & CIRRELT (Interuniversity Research Centre on Enterprise Networks, Logistics and Transportation)

Standing: GPA 4.0/4.0 – Distinction (Honors)

Engineering Degree, Grande École Program | Arts et Métiers ParisTech (ENSAM)

France, 2022-2026

A top generalist engineering Grande École in France for 250 years.
Standing: Top 10% – Recipient of an international study scholarship from the alumni society

Quantitative core: applied mathematics (Fourier analysis, symbolic and numerical computation, optimization), programming and databases, automation and control, embedded systems.

Engineering core: mechanical and process engineering, materials, energy, industrial engineering (production management, Lean, Six Sigma), economics and accounting.

Preparatory Classes for the Grandes Écoles (CPGE), PTSI/PT* | Lycée Livet

France, 2020-2022

Preparation for the top French engineering schools entrance exams
Weekly coursework: 9h mathematics, 8.5h engineering science, 8h physics, 2h computer science, 4h languages, 4h written exams, 2h oral exams

SKILLS

Programming: Python (pandas, NumPy, scikit-learn, XGBoost, PyTorch), SQL, VBA, MATLAB, G-code, C/C++ (embedded)

Machine Learning: Supervised learning and gradient boosting, Learning-to-Rank, neural networks with human feedback, NLP and information extraction, time series, unsupervised learning

LLM systems: Retrieval-augmented generation (hierarchical RAG), Model Context Protocol (MCP) servers, local LLM deployment, prompt and context engineering

Data & cloud: Azure (ADLS Gen2, Functions, App Service, Containers), SQLite, Docker, Git, Streamlit, API integration and web scraping, Excel/VBA automation

Operations research: Mathematical optimization, convex and stochastic optimization, Monte Carlo simulation, numerical simulation

Languages: French (native), English (fluent, C1 - TOEIC 950), Spanish (basic)

Certifications: IBM Data Science Specialization (160h, Coursera), Stanford Machine Learning Specialization (80h, Coursera)

Open online courses: Stanford CS229 Machine Learning, Stanford CS224n NLP with Deep Learning, Stanford CS230 Deep Learning

LEADERSHIP

Vice-President of the School's History Association | ENSAM

France, 2023-2024

Association responsible for preserving the school's history, traditions, and buildings.
■ Created an annotated virtual-tour website; awarded a €1,000 FSDIE grant.